

Software Test Plan (STP) - Online Bookstore

Project: Online Bookstore (Team 2, Section C)

Version: 1.0

Authors: Deeptha S (QA Lead), Hardhick M Gowda (CI & Automation Lead), Chenchu Sreekrishna Koushik, G H Pramod

Date: 05-10-2026

Status: Baselined for Sprint 1

1. Introduction

Purpose: This plan explains how we will test the Online Bookstore v1.0 - what we test, how, when, and who does it.

Scope: Login and accounts, catalogue and search, cart and orders, payments (sandbox), seller tools, and admin tools. The internals of outside services (payment gateway, e-mail provider, MongoDB Atlas) are not tested.

References: SRS v1.1, SAD v1.0, IEEE 829.

Definitions: TC - test case. STC - system test case (a full user journey, SRS Appendix C.3). RTM - Requirement Traceability Matrix. Stub - a fake version of an outside service used in tests.

2. Test Items

- Website (React)
- Server: Auth & User, Catalogue & Search, Cart/Order/Payment, Seller/Admin/Reporting services
- Database models and the e-mail, cache and payment adapters
- CI pipeline and deployment

3. Features to be Tested

All requirements from SRS v1.1 are tested. The test case for each one is listed below; full test cases are in section 13.

Feature group	Requirements	Test cases
Accounts and access	FR-01 to FR-06	TC-A-01 to TC-A-06
Catalogue, search, reviews	FR-07 to FR-12	TC-C-01 to TC-C-06
Cart, checkout, payment, orders	FR-13 to FR-18	TC-O-01 to TC-O-06

Feature group	Requirements	Test cases
Seller and admin	FR-19 to FR-24	TC-S-01 to TC-S-06
Performance, availability, usability, quality	NFR-01 to NFR-08	TC-N-01 to TC-N-08
Security	SR-01 to SR-12	TC-SEC-01 to TC-SEC-10

4. Features Not to be Tested

- Payment gateway internals and real-money payments (sandbox only).
- E-mail provider delivery and MongoDB Atlas backups (managed by the providers).
- Courier delivery, mobile apps and old browsers (out of scope for v1.0).

5. Test Approach / Strategy

Levels:

Level	What	Tool
Unit	Single functions, e.g. ISBN check, coupon maths	Jest
Integration (API)	Each API endpoint with a test database	Jest + Supertest
System	Full user journeys on the staging site (STC-01 to STC-16)	Playwright, manual
Acceptance	Demo scenarios reviewed by the team and faculty	Manual

Types: functional, regression (on every pull request), performance (k6), usability, cross-browser and security.

Entry criteria: build passes CI; staging is up; test data loaded.

Exit criteria: all planned tests run; no critical bugs open; at least 70 per cent test coverage of service code.

Bugs: logged as GitHub Issues with severity (Critical, Major, Minor, Cosmetic) and the failing test ID.

5.1 Security Validation

Security tests check the security requirements SR-01 to SR-12 from SRS section 6.3, against the threats listed in SAD section 3.9.

Test ID	Requirement	What we check	Pass if
TC-SEC-01	SR-01	Passwords hashed and never logged	Only bcrypt hashes; nothing in logs
TC-SEC-02	SR-02	HTTPS only (TLS 1.2+), HTTP redirected	Old TLS refused; HTTP gets 301
TC-SEC-03	SR-03	No card data on our server	Card number not in database, logs or our API traffic

Test ID	Requirement	What we check	Pass if
TC-SEC-04	SR-04	Token expiry and refresh reuse	Expired token refused; reused refresh token signs the user out everywhere
TC-SEC-05	SR-05, SR-06	Access control (role and owner)	Wrong role gets 403; other users' data gets 404
TC-SEC-06	SR-07	Injection and XSS attempts	Attacks rejected or shown as plain text; ZAP finds no high-risk issues
TC-SEC-07	SR-08	Fake or repeated payment webhook	Bad signature refused; repeated message changes nothing
TC-SEC-08	SR-09	Audit log cannot be changed	No way to edit or delete entries
TC-SEC-09	SR-10	Too many logins or OTP requests	6th failed login gets 429; 4th OTP request in an hour gets 429
TC-SEC-10	SR-11, SR-12	Secrets and libraries check	No secrets in Git; no high-risk npm packages; OWASP Top 10 checklist done

Tools: OWASP ZAP, testssl.sh, gitleaks, npm audit, Postman.

6. Test Environment

- **Local:** Node.js 20, MongoDB 7 (or in-memory MongoDB for tests).
- **CI:** GitHub Actions runs lint, tests and coverage on every pull request.
- **Staging:** deployed server and website, MongoDB Atlas, Redis, payment gateway in sandbox mode, test e-mail inbox.
- **Browsers:** latest Chrome, Edge, Firefox and Safari; a 360 px phone view.
- **Test data:** seed script with 12 test accounts and 10,000 book listings.

7. Test Schedule

Milestone	Dates
Test plan submitted	07-10-2026
Sprint 1 testing (accounts, catalogue)	05-10-2026 to 18-10-2026
Sprint 2 testing (checkout, payment, listings)	19-10-2026 to 01-11-2026
Sprint 3 testing (orders, seller, admin)	02-11-2026 to 15-11-2026
Final system and security testing	16-11-2026 to 22-11-2026
Demo and acceptance	Week of 23-11-2026 (to be confirmed)

8. Test Deliverables

- This test plan with test cases
- Automated tests in the repository
- CI reports, bug reports, load-test and security-scan results

- Test summary report at the end of each sprint

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Deeptha S	Owns this plan; runs system, performance and cross-browser tests; writes test reports
CI & Automation Lead	Hardhick M Gowda	Owns the CI pipeline, staging environment and automated regression suite; tests seller and admin features
Security Test Lead	Chenchu Sreekrishna Koushik	Runs security tests (section 5.1); tests account features
Test Engineer	G H Pramod	Tests cart, checkout, payment and order features
Developers	All four members	Write unit and API tests for their own code; fix their bugs

10. Risks and Mitigation

Risk	Mitigation
Payment sandbox is down	Use a stub gateway in tests
Features arrive late	Write tests alongside the code in each sprint
Free-tier limits slow the load test	Keep load tests short; use a test e-mail inbox
Developers test their own code	Each system test is also run by another team member

11. Assumptions & Dependencies

- Payment sandbox and free-tier services stay available.
- Features reach staging on the sprint dates.

12. Suspension & Resumption Criteria

Suspend if staging is down for more than 4 hours, or if more than 30 per cent of tests are blocked by one bug. **Resume** when the bug is fixed and the smoke test (log in, search, add to cart) passes.

13. Test Case Management & Traceability

Every requirement has at least one test case, and automated tests use the test ID in their name (for example "TC-A-01 ...") so results map straight to the RTM in SRS Appendix C.

Requirement	Test case	System test
FR-01	TC-A-01	STC-01

Requirement	Test case	System test
FR-02	TC-A-02	STC-01
FR-03	TC-A-03	STC-02
FR-04	TC-A-04	STC-02
FR-05	TC-A-05	STC-03
FR-06	TC-A-06	STC-03
FR-07	TC-C-01	STC-04
FR-08	TC-C-02	STC-04
FR-09	TC-C-03	STC-05
FR-10	TC-C-04	STC-05
FR-11	TC-C-05	STC-06
FR-12	TC-C-06	STC-06
FR-13	TC-O-01	STC-07
FR-14	TC-O-02	STC-07
FR-15	TC-O-03	STC-08
FR-16	TC-O-04	STC-08
FR-17	TC-O-05	STC-09
FR-18	TC-O-06	STC-09
FR-19	TC-S-01	STC-10
FR-20	TC-S-02	STC-10
FR-21	TC-S-03	STC-11
FR-22	TC-S-04	STC-11
FR-23	TC-S-05	STC-12
FR-24	TC-S-06	STC-12
NFR-01	TC-N-01	STC-13
NFR-02	TC-N-02	STC-13
NFR-03	TC-N-03	STC-14
NFR-04	TC-N-04	STC-14
NFR-05	TC-N-05	STC-15
NFR-06	TC-N-06	STC-15
NFR-07	TC-N-07	STC-16
NFR-08	TC-N-08	STC-16
SR-01 to SR-12	TC-SEC-01 to TC-SEC-10	STC-14

Test cases

32 functional and non-functional test cases are listed below; the 10 security test cases are in section 5.1. Status for all is "Not run" until Sprint 1 testing starts.

ID	Req	Scenario	Steps	Expected result
TC-A-01	FR-01	Register a new buyer	Register with valid details; register again with the same e-mail; try a weak password	Account created (201); duplicate e-mail refused (409); weak password refused (400)
TC-A-02	FR-02	Verify e-mail with OTP	Log in before verifying; enter a wrong OTP 3 times; enter the right OTP on a new code	Login blocked until verified; locked after 3 wrong tries; right code activates account
TC-A-03	FR-03	Log in and log out	Log in; refresh the token; log out; try the old refresh token	Tokens issued (15 min / 7 days); refresh works; after logout the old token is refused
TC-A-04	FR-04	Reset a forgotten password	Request reset for a known and an unknown e-mail; use the link twice	Same message for both e-mails; link works once; old sessions signed out
TC-A-05	FR-05	Manage addresses	Add two addresses marked default; add a 5-digit PIN code	Only one default address; invalid PIN code refused
TC-A-06	FR-06	Role-based access	Buyer opens a seller page; blocked user logs in	Buyer gets 403 and it is logged; blocked user refused
TC-C-01	FR-07	Browse the catalogue	Open pages 1, 2 and 3 with 45 books seeded	20, 20 and 5 books shown with title, author, price and rating
TC-C-02	FR-08	Search by keyword	Search "wings", "kalam", an ISBN, and "zzqx"	Correct books found, best match first; "zzqx" shows no results and suggests a category
TC-C-03	FR-09	Filter and sort	Filter by category, condition, price and rating; sort by price	Only matching books shown, in the right order
TC-C-04	FR-10	View book details	Open a book with 4 listings (2 active)	Only the 2 active offers shown, cheapest first
TC-C-05	FR-11	Write a review	Review a book not yet delivered; review a delivered book twice	First refused (403); second accepted once, then refused (409)
TC-C-06	FR-12	Related titles	Open a book whose author has other books	Up to 6 related books, same author first, not the book itself
TC-O-01	FR-13	Use the cart	Add a book with stock 3 and set quantity 5; log out and back in	Quantity capped at 3; cart is still there after login
TC-O-02	FR-14	Use the wishlist	Save a book; move it to the cart	Book moves to cart and leaves the wishlist

ID	Req	Scenario	Steps	Expected result
TC-O-03	FR-15	Checkout with a coupon	Cart of Rs 450 with coupon SAVE50; then a cart of Rs 520	Payable Rs 440 (450 + 40 shipping - 50); shipping free at Rs 500 or more
TC-O-04	FR-16	Pay for an order	Send a failed payment, then a successful one	Failed: order stays PLACED; success: order CONFIRMED and stock reduced
TC-O-05	FR-17	Get the invoice	Download the invoice of a confirmed order	PDF with order number, items and totals; also e-mailed
TC-O-06	FR-18	Cancel or return	Cancel a confirmed order; try to cancel a shipped order; return after 8 days	First cancelled and stock restored; second and third refused
TC-S-01	FR-19	Create a listing	Enter a known ISBN; enter an invalid ISBN	Details auto-filled and listing created; invalid ISBN refused
TC-S-02	FR-20	Manage stock and price	Set price above MRP; set stock to 5; set stock to 0	Price refused; low-stock alert sent; listing becomes OUT_OF_STOCK
TC-S-03	FR-21	Ship an order	Accept, pack, and ship with courier and tracking number	Shipment saved; buyer notified at each step
TC-S-04	FR-22	Seller dashboard	Open dashboard for a month with known sales	Totals, commission and settlement match the orders
TC-S-05	FR-23	Admin moderation	Block a listing without a reason, then with a reason	Refused without reason; with reason the listing disappears and the action is logged
TC-S-06	FR-24	Monthly analytics	Export the monthly report as CSV and PDF	Figures match the database; finishes within 10 s
TC-N-01	NFR-01	Search speed	Load test with 200 users and 10,000 listings (k6)	95 per cent of searches return within 2 s
TC-N-02	NFR-02	Availability	Uptime monitor checks /health every 5 minutes	Uptime of 99 per cent or more
TC-N-03	NFR-03	Password storage	Register, then check the database and logs	Only bcrypt (cost 12) hashes stored; no plain password anywhere
TC-N-04	NFR-04	Usability	Buy a book on a 360 px screen using only the keyboard	Done in 4 screens or fewer; everything usable
TC-N-05	NFR-05	No overselling	Two buyers pay for the last copy at the same moment	Only one order succeeds; stock never goes below 0
TC-N-06	NFR-06	Code quality gate	Open a pull request with a lint error	CI fails and the merge is blocked
TC-N-07	NFR-07	Browser support	Run the main flows in Chrome, Edge, Firefox and Safari	All flows work in every browser

ID	Req	Scenario	Steps	Expected result
TC-N-08	NFR-08	Audit log	Do admin and payment actions, then check the log	Every action logged with who, what and when; entries cannot be edited

14. Test Metrics & Reporting

- Percentage of test cases run and passed
- Number of open bugs by severity
- Requirement coverage (target: 100 per cent)
- Code coverage of service code (target: at least 70 per cent)

The QA Lead shares a short test report at each sprint review.

15. Approvals

Role	Name	Signature / Date
QA Lead	Deeptha S	
CI & Automation Lead	Hardhick M Gowda	
Security Test Lead	Chenchu Sreekrishna Koushik	
Test Engineer	G H Pramod	